Practical no. 5

Aim

To configure a DNS Server and Web Server using Cisco Packet Tracer.

Objective

- To understand the working of DNS (Domain Name System).
- To configure a Web Server and host a website.
- To map domain names to IP addresses using a DNS Server.
- To test connectivity between client and server.

Theory

DNS (Domain Name System) is used to translate domain names into IP addresses. This helps users access websites without remembering numerical IP addresses.

A Web Server is used to host websites and deliver web pages to clients using HTTP protocol.

In Cisco Packet Tracer:

- The DNS Server resolves domain names.
- The Web Server hosts the website.
- The Client (PC) accesses the website using a browser.

Requirements

- Cisco Packet Tracer software
- 1 Server (for DNS & Web)
- 1 PC (Client)
- Switch
- Connecting cables

Network Topology

Procedure

Step 1: Create Network

- Open Cisco Packet Tracer.
- Add:
  - 1 Switch
  - 1 Server
  - 1 PC
- Connect all devices using Copper Straight-through cables.

Step 2: Configure IP Addressing

Server Configuration:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

PC Configuration:

- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0
- DNS Server: 192.168.1.2

Step 3: Configure DNS Server

- Click on Server → Services → DNS
- Turn DNS service **ON**
- Add entry:
  - Name: www.mywebsite.com
  - Address: 192.168.1.2

Step 4: Configure Web Server

- Go to Server → Services → HTTP
- Turn HTTP service **ON**
- Edit index.html page:

```
<html>
  <body>
    <h1>Welcome to My Website</h1>
  </body>
</html>
```

Step 5: Test the Configuration

- Open PC → Desktop → Web Browser
- Type: www.mywebsite.com
- Press Enter

Output

- The webpage should display:
  “Welcome to My Website”

Result

The DNS Server and Web Server were successfully configured in Cisco Packet Tracer. The client was able to access the website using the domain name.

Conclusion

- DNS simplifies network usage by converting domain names to IP addresses.
- Web Servers host and deliver web pages.
- Cisco Packet Tracer helps simulate real-world networking concepts effectively.

Practical no. 6

Aim

To configure a DHCP Server using Cisco Packet Tracer.

Objective

- To understand the working of DHCP (Dynamic Host Configuration Protocol).
- To automatically assign IP addresses to client devices.
- To reduce manual IP configuration errors.
- To test dynamic IP allocation in a network.

Theory

DHCP (Dynamic Host Configuration Protocol) is a network management protocol used to automatically assign IP addresses, subnet masks, default gateways, and DNS servers to devices on a network.

Instead of manually configuring IP addresses for each device, DHCP allows a server to distribute IP configurations dynamically from a predefined pool.

Working of DHCP (DORA Process):

- **Discover** – Client sends request for IP
- **Offer** – DHCP server offers an IP address
- **Request** – Client requests the offered IP
- **Acknowledge** – Server confirms assignment

Requirements

- Cisco Packet Tracer software
- 1 Server (DHCP Server)
- Multiple PCs (Clients)
- Switch

- Connecting cables

IP Configuration (Example Setup)

Server (DHCP):

- IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0

DHCP Pool Settings:

- Start IP: 192.168.1.100
- End IP: 192.168.1.110
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.1

Clients (PCs):

- Set to **DHCP (Automatic IP)**

Advantages of DHCP

- Automatically assigns IP addresses to devices
- Reduces manual configuration effort
- Minimizes IP conflicts (duplicate IP issues)
- Efficient management of large networks
- Centralized control of network configuration

Disadvantages of DHCP

- DHCP server failure can stop network connectivity
- Security risks (unauthorized devices may get IPs)
- Requires initial setup and maintenance
- Not suitable for devices needing static IPs (like servers)

Output

- PCs receive IP addresses automatically (e.g., 192.168.1.100, 101, etc.)

Result

The DHCP Server was successfully configured in Cisco Packet Tracer, and IP addresses were automatically assigned to client devices.

Conclusion

DHCP simplifies network administration by automating IP address allocation and reducing configuration errors, making it essential for modern networks.

Practical no. 7

Aim

To configure an Email Server using Cisco Packet Tracer.

Objective

- To understand the working of an Email Server.
- To configure email services (SMTP & POP3).
- To create and manage user email accounts.
- To send and receive emails between clients.

Theory

An Email Server is used to send and receive electronic mails over a network.

It mainly uses two protocols:

- **SMTP (Simple Mail Transfer Protocol)** – used to send emails.
- **POP3 (Post Office Protocol)** – used to receive emails.

In Cisco Packet Tracer:

- The Server acts as a Mail Server.
- Users are created with email IDs and passwords.
- Clients (PCs) use email software to communicate through the server.

Requirements

- Cisco Packet Tracer software
- 1 Server (Email Server)
- 2 PCs (Clients)
- Switch
- Connecting cables

IP Configuration (Example Setup)

Server:

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

PC1:

- IP Address: 192.168.1.3
- Subnet Mask: 255.255.255.0

PC2:

- IP Address: 192.168.1.4
- Subnet Mask: 255.255.255.0

Email Server Configuration (Summary)

- Enable **SMTP** and **POP3** services on the server
- Set domain name: `mail.com`
- Create user accounts:
  - user1 → user1@mail.com
  - user2 → user2@mail.com
- Configure Email on PCs:
  - Email ID: user1@mail.com / user2@mail.com
  - Incoming Mail Server (POP3): 192.168.1.2
  - Outgoing Mail Server (SMTP): 192.168.1.2

Advantages of Email Server

- Fast and reliable communication
- Allows file sharing via attachments
- Centralized user management
- Accessible from multiple devices
- Cost-effective communication system

Disadvantages of Email Server

- Requires proper configuration and maintenance

- Vulnerable to spam and security threats
- Server failure can disrupt communication
- Storage limitations for emails
- Needs network connectivity

Output

- Email successfully sent from one PC to another
- Message received in inbox

Result

The Email Server was successfully configured in Cisco Packet Tracer, and emails were sent and received between clients.

Conclusion

Email servers enable efficient communication over networks using SMTP and POP3 protocols, and Cisco Packet Tracer helps simulate and understand their working.

Practical no. 8

Aim

To configure IP Routing using RIP (Routing Information Protocol) in Cisco Packet Tracer.

Objective

- To understand dynamic routing using RIP.
- To configure routing between multiple networks.
- To enable communication between different subnets.
- To learn how routers exchange routing information.

Theory

RIP (Routing Information Protocol) is a dynamic routing protocol used to share routing information between routers.

- It uses **hop count** as a metric (maximum 15 hops).
- Works on the **Distance Vector algorithm**.
- Routers periodically exchange routing tables.
- Uses **UDP port 520**.

Versions of RIP:

- **RIP v1** – Classful (no subnet mask support)
- **RIP v2** – Classless (supports subnetting and VLSM)

Requirements

- Cisco Packet Tracer software
- 2 Routers
- 2 Switches
- 2 PCs
- Connecting cables

IP Configuration (Example Setup)

Network 1:

- PC1: 192.168.1.2
- Router1 Interface: 192.168.1.1

Network 2:

- PC2: 192.168.2.2
- Router2 Interface: 192.168.2.1

Router-to-Router Link:

- Router1: 10.0.0.1
- Router2: 10.0.0.2

RIP Configuration Commands

Router 1 Configuration

```
enable
configure terminal
router rip
version 2
network 192.168.1.0
network 10.0.0.0
exit
```

Router 2 Configuration

```
enable
configure terminal
router rip
version 2
network 192.168.2.0
network 10.0.0.0
exit
```

Advantages of RIP

- Simple and easy to configure
- Suitable for small networks

- Automatically updates routing tables
- No need for manual route configuration

Disadvantages of RIP

- Limited to 15 hops (not for large networks)
- Slow convergence time
- Can cause network traffic due to periodic updates
- Less secure compared to advanced protocols

Output

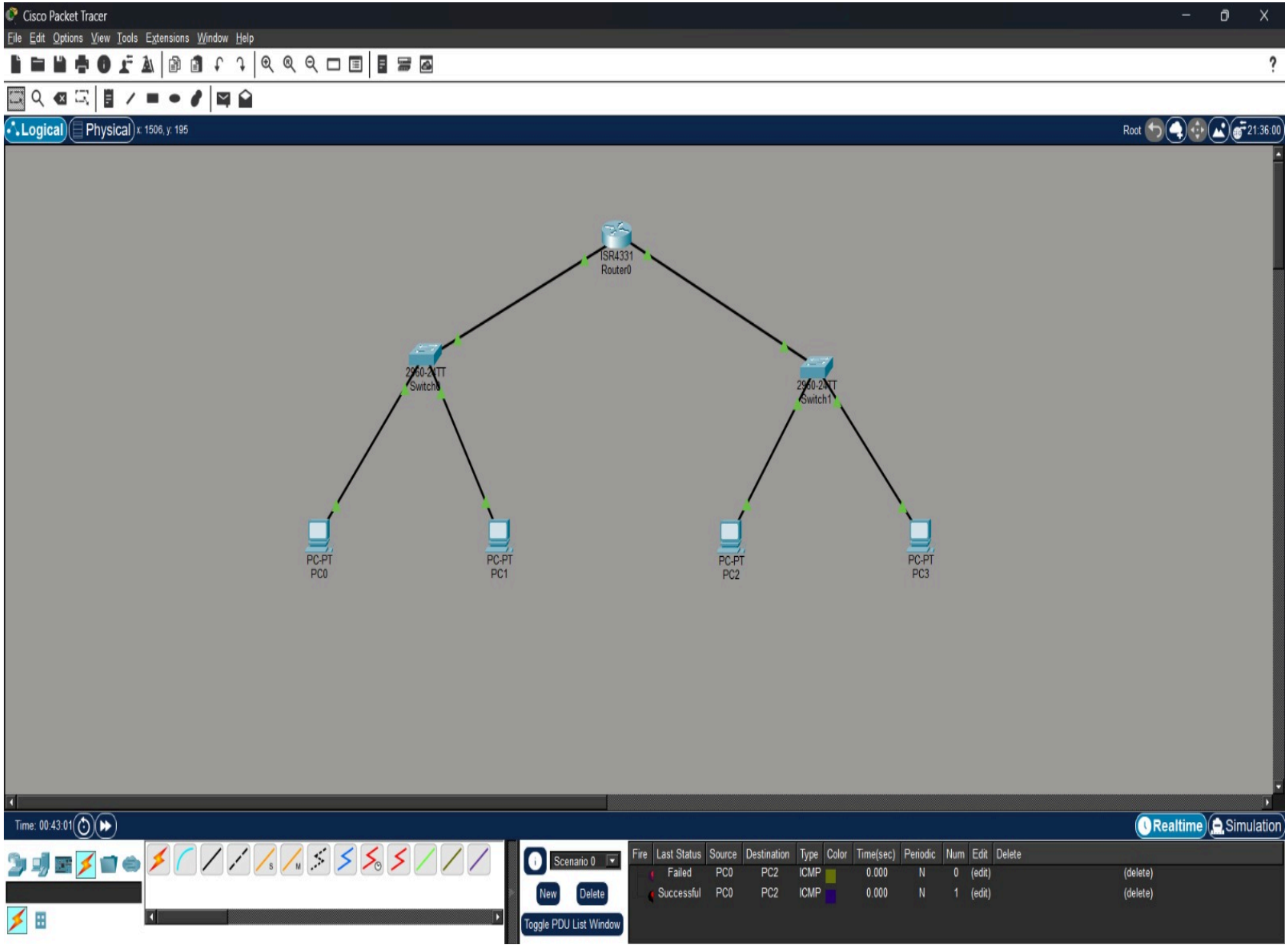
- Successful **ping** between PC1 and PC2
- Routing tables updated dynamically

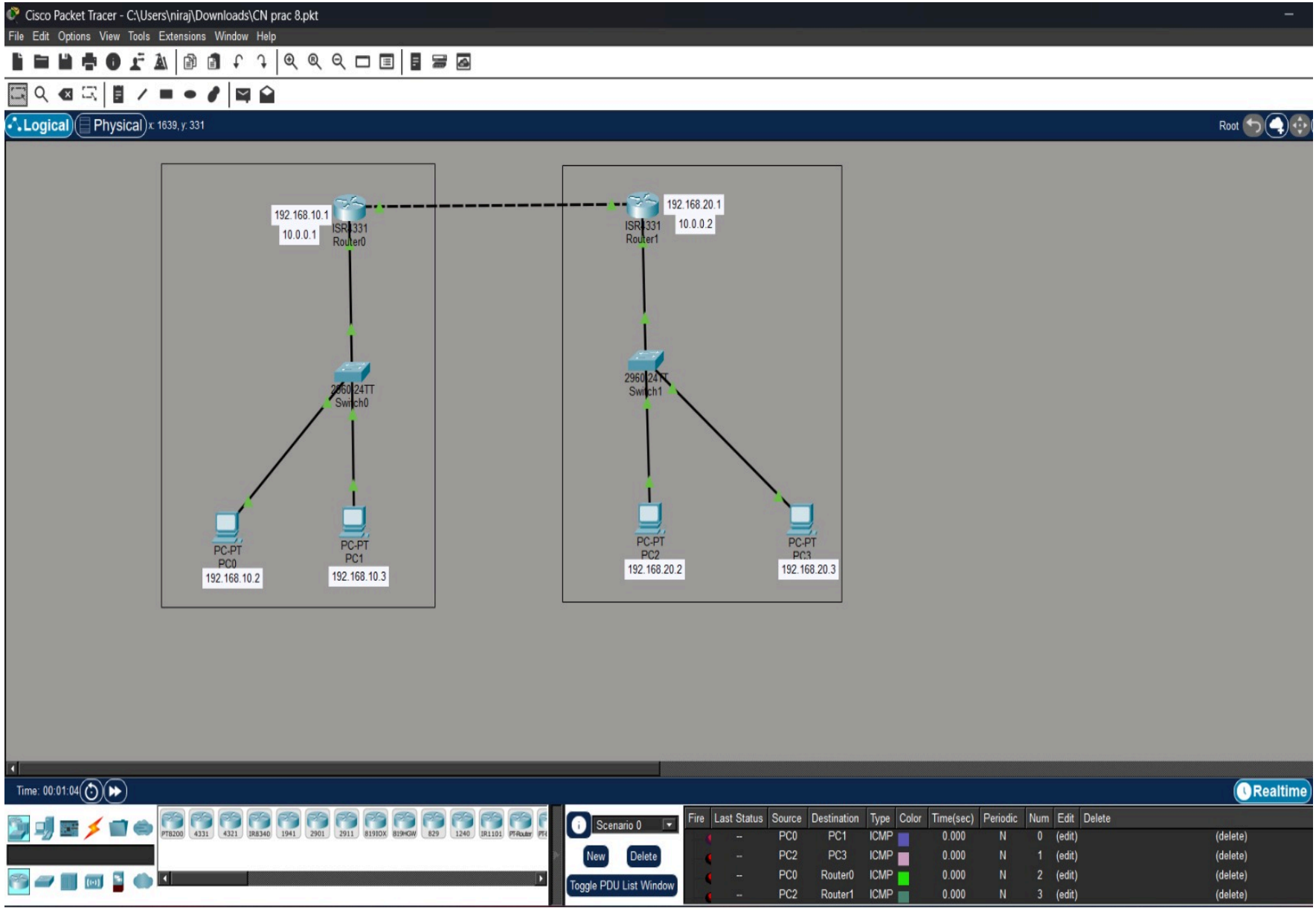
Result

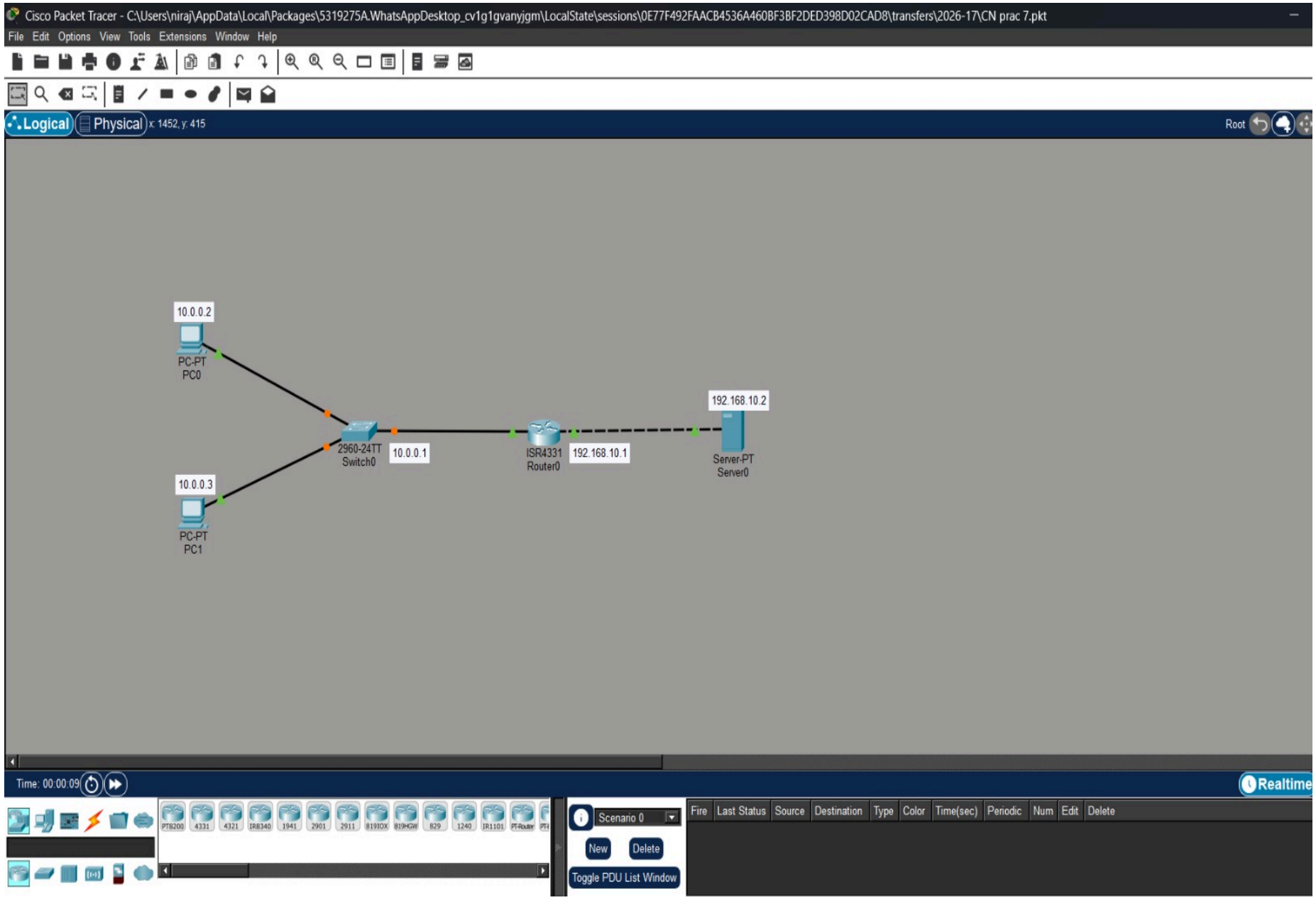
IP routing using RIP was successfully configured, and communication between different networks was achieved.

Conclusion

RIP is a basic dynamic routing protocol that helps routers exchange routing information automatically, making it useful for small-scale networks.


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1506, y: 195
Root
21:36:00
ISR4331
Router0
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Time: 00:43:01
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Failed PC0 PC2 ICMP 0.000 N 0 (edit) (delete)
Successful PC0 PC2 ICMP 0.000 N 1 (edit) (delete)


Cisco Packet Tracer - C:\Users\niraj\Downloads\CN prac 8.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1639, y: 331
Root
192.168.10.1
10.0.0.1
ISR4331
Router0
192.168.20.1
10.0.0.2
ISR4331
Router1
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
192.168.10.2
PC-PT
PC1
192.168.10.3
PC-PT
PC2
192.168.20.2
PC-PT
PC3
192.168.20.3
Time: 00:01:04
Realtime
Scenario 0
New
Delete
Toggle PDU List Window
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete
-- | PC0 | PC1 | ICMP | 0.000 | N | 0 | (edit) | (delete)
-- | PC2 | PC3 | ICMP | 0.000 | N | 1 | (edit) | (delete)
-- | PC0 | Router0 | ICMP | 0.000 | N | 2 | (edit) | (delete)
-- | PC2 | Router1 | ICMP | 0.000 | N | 3 | (edit) | (delete)


Cisco Packet Tracer - C:\Users\niraj\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\0E77F492FAACB4536A460BF3BF2DED398D02CAD8\transfers\2026-17\CN prac 7.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1452, y: 415
Root
10.0.0.2
PC-PT
PC0
10.0.0.3
PC-PT
PC1
2960-24TT
Switch0
10.0.0.1
ISR4331
Router0
192.168.10.1
192.168.10.2
Server-PT
Server0
Time: 00:00:09
Realtime
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
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Type
Color
Time(sec)
Periodic
Num
Edit
Delete